1. Security Camera

- **Descriptive Analysis (What happened?)**
    - **Metric Tracked:** A 25% spike in customer support tickets regarding "motion detection false alarms" and a concurrent 12% drop in app daily active users (DAU) over the last quarter.
- **Diagnostic Analysis (Why did it happen?)**
    - **Root Cause:** A recent firmware update recalibrated the computer vision model to increase sensitivity, causing pets, shadows, and swaying tree branches to trigger high-priority push notifications, leading to user fatigue and disabled alerts.
- **Predictive Analysis (What might happen next?)**
    - **Forecast:** If left unaddressed, customer churn at the upcoming annual subscription renewal window is projected to increase by 8% over the next 60 days.
- **Prescriptive Analysis (What should we do?)**
    - **Action Plan:** Roll out a hotfix update introducing customizable motion zones and adjustable AI sensitivity sliders within the mobile app. Send an email campaign to affected users explaining the fix and offering a guided setup tutorial.

2. Wearable Fitness Tracker

- **Descriptive Analysis (What happened?)**
    - **Metric Tracked:** User engagement with the advanced sleep-tracking feature dropped by 35% among new subscribers within their first 14 days of activation.
- **Diagnostic Analysis (Why did it happen?)**
    - **Root Cause:** User journey funnel analysis revealed a high friction point during device synchronization; users struggled with Bluetooth pairing on specific Android models, preventing them from accessing baseline health metrics.
- **Predictive Analysis (What might happen next?)**
    - **Forecast:** Users who fail to sync their device within the first 3 days have a 70% higher likelihood of canceling their subscription before the free trial ends.
- **Prescriptive Analysis (What should we do?)**
    - **Action Plan:** Redesign the onboarding UI to feature an in-app troubleshooting wizard for Bluetooth connectivity. Trigger automated push notifications with step-by-step pairing videos if synchronization is not detected within 24 hours of unboxing.

3. Plant-Based Meal Replacement Shake

- **Descriptive Analysis (What happened?)**
    - **Metric Tracked:** E-commerce repeat purchase rates for the "Vanilla Bean" flavor fell by 20%, while repeat purchases for "Chocolate Fudge" remained steady.
- **Diagnostic Analysis (Why did it happen?)**
    - **Root Cause:** Textural feedback scraped from recent customer reviews and return logs indicated that a recent change in plant protein suppliers resulted in a chalky consistency for the vanilla batch.
- **Predictive Analysis (What might happen next?)**
    - **Forecast:** Inventory turnover for Vanilla Bean stock will slow down by 40%, leading to excess warehouse storage costs and potential stock expiration within the quarter.
- **Prescriptive Analysis (What should we do?)**
    - **Action Plan:** Pause production of the current vanilla formulation, revert to the previous protein supplier specification, and offer affected recent buyers a discount voucher on a different flavor.

4. On-Demand Video Streaming Service

- **Descriptive Analysis (What happened?)**
    - **Metric Tracked:** Monthly subscriber churn rose by 14% in the mid-tier ad-supported plan, alongside a 22% drop in total hours watched per user over 30 days.
- **Diagnostic Analysis (Why did it happen?)**
    - **Root Cause:** Video player analytics showed that a recent change to the ad insertion algorithm increased unskippable ad frequency during high-climax scene transitions, causing high user drop-off rates mid-episode.
- **Predictive Analysis (What might happen next?)**
    - **Forecast:** Subscription cancellations will increase by another 10% next month as users reach the end of their current billing cycles without finishing active shows.
- **Prescriptive Analysis (What should we do?)**
    - **Action Plan:** Cap mid-content ad breaks to natural scene transitions, limit unskippable ads to a maximum of 30 seconds per break, and push a personalized "Resume Watching" notification with an ad-free trial preview to lapsed subscribers.

5. Automated Coffee Machine

- **Descriptive Analysis (What happened?)**
    - **Metric Tracked:** Maintenance request tickets for internal pump blockages surged by 35% across machines deployed in hard-water regions.
- **Diagnostic Analysis (Why did it happen?)**
    - **Root Cause:** Diagnostic logs indicated that the machine's default descaling reminder algorithm relied solely on brew cycles rather than adjusting for regional water hardness parameters.
- **Predictive Analysis (What might happen next?)**
    - **Forecast:** Pump replacements will increase by 50% in affected cities over the next 6 months, driving up overall service operation expenses.
- **Prescriptive Analysis (What should we do?)**
    - **Action Plan:** Update the companion app to prompt users for local water hardness settings during initial setup, recalibrating the descaling alert frequency dynamically based on regional water hardness data.